

This PhD thesis in theoretical physics takes on a somewhat atypical form. It consists of a long introductory part (chapters 1-4, about 90 pages), after which the 6 journal publications of the candidate are reproduced (each with a short summary and the contributions of the candidate specified).

Reading this thesis is not an easy task: it is a combination of cutting-edge physics with highly abstract mathematics (category theory). This is further complicated by the fact that all familiar (for me as a mathematician) mathematical terms take on additional physical names. An object in a category becomes a defect, sector or charge depending on context. Nevertheless I could really appreciate the efforts made in e.g. chapter 3 to explain both the mathematics and physics.

Concerning the 6 papers that form the bulk of the thesis:

- Chapter 5 is a long invitation to Tensor Networks, based on lectures by one of the supervisors. Although written by the candidate, it seems that the selection of the material, presentation and organization was mainly due to the supervisor.
- Chapter 6 concerns a study of phases in one dimensional systems when a frieze group symmetry is present. An interesting link with group cohomology is uncovered.
- In chapter 7 a map (in the form of a 'quantum circuit') is constructed between two string-nets (unitary fusion categories). The mapping seems very explicit.
- Chapter 8 provides an equivalence between gauging and duality transformations for 1D quantum lattice models, again using categorical techniques.
- Chapter 9 considers abelian group symmetry and constructs a general procedure to lift the symmetry to the boundary of a one dimension higher model.
- Chapter 10 finally investigates a type of duality using higher categories.

In conclusion this is a high level thesis on (very) modern physics, with a deep mathematical flavor. It contains many new results and is executed very well.

Remarks and questions:

- Although there are 7 frieze groups, up to group isomorphism there are only 4 distinct ones. Does this show up in your results?
There are also 17 wallpaper groups (in 2D). Could a similar construction be done for them?
- If AI or LLMs were used in the research or writing process, this should be disclosed in the thesis.
- The own contributions of the candidate seem somewhat understated in the way they are explained in chapter 5 - 10.
E.g. in Chapter 7 he states that his contribution was the explicit derivation of the map, which in fact seems to be the crucial development the paper hinges on.